



RAPID PROTOTYPING METHODOLOGIES FOR COMMERCIAL APPLICATION

Overview

Rapid prototyping serves various purposes, with cost-effectiveness and a decreased rate of product failure being the primary advantages. As the world emerges from recent economic disruptions, the evolving scope of product development presents unique opportunities for product professionals and business leaders alike, with rapid prototyping emerging at the top.

The MIT xPRO **Rapid Prototyping Methodologies for Commercial Application** program imparts an intimate understanding of product performance, fewer design iterations, and enhanced early understanding of scaling and manufacturability issues. MIT developed the plastic face shield, an integral part of PPE suits during COVID-19, and generated three million face shields in less than 30 days for 4,000 frontline facilities fighting the virus.

Faculty member Martin Culpepper, professor of mechanical engineering and director of Project Manus, MIT's research and development office dedicated to helping create and expand access to technology, led the mass production process with his expertise in rapid prototyping and 3D cutting and printing. Culpepper described their technique as one that "combines low-cost materials with high-rate manufacturing," a description that embodies rapid prototyping as a whole.

This program offers product engineers, managers, designers, business leaders, and functional heads a cutting-edge perspective on the use of rapid prototyping technology, paired with the knowledge needed to streamline product design and manufacturing. Participants will come away with a thorough understanding of different types of fabrication processes and the ability to discern the best strategy for building and evaluating a prototype and ultimately giving their organization an advantage in the go-to-market process.

PRICE

USD 2,600

DURATION

8 weeks
4-6 hours per week

The rapid prototyping service market is expected to grow at a 16.7% CAGR during the forecast period, reaching USD 1,511.2 million.

The automotive industry, consumer goods, aerospace and defense, and medical devices employ rapid prototyping extensively.

The rapid prototyping market will reach USD 1 billion by 2030.

(Source: Global Rapid Prototyping Market Industry Analysis, Share Report till 2026)

Program Highlights



A certificate and four Continuing Education Units (CEUs) from MIT xPRO



Insights and case studies from renowned MIT faculty



A capstone project where participants will make fabrication decisions for a prototype, assess test results, optimize cost value, and develop a presentation

Services Offered by Emeritus



Insights and case studies from renowned MIT faculty



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Why This Program

- 📦 You will learn from a leading MIT expert in rapid prototyping – Professor Martin Culpepper.
- 📦 The final module features four guest speakers who teach at MIT and will share insights on the application of rapid prototyping in healthcare, medical devices, and research and development, along with other uses.
- 📦 There is one live session by the faculty and weekly office hours with learning facilitators.
- 📦 Every module uses MIT's Project Manus (COVID-19) face shield to explain the application of rapid prototyping technology.
- 📦 The program experience includes recorded demonstration videos about the prototyping process.
- 📦 Capstone project – Module 5 onwards will apply the learnings of the program.

Who Is This Program For?



Product engineers who:

Capstone project – Module 5 onwards will apply the learnings of the program.



Product managers and designers who want to:

Use prototyping to refine product concepts and make design and manufacturing decisions that create efficiencies in the go-to-market process



Product managers and designers who want to:

Use prototyping to refine product concepts and optimize resource allocation

Key Takeaways

- Identify rapid prototyping fundamentals and the various techniques for developing the prototypes.
- Recognize how rapid prototyping can increase agility in design and manufacturing.
- Understand how to map desired product attributes to concept prototype attributes.
- Analyze the benefits and drawbacks of serial and parallel fabrication processes to develop a rapid prototype.
- Make fabrication decisions to define strategies for creating and assessing the effectiveness of a prototype.
- Assess data gathered from concept prototypes to make smart decisions on developing your desired product.
- Identify how to assess and optimize the cost value of a rapid prototype.
- Determine the future trends and technologies used to enhance rapid prototyping.



Program Curriculum

Module 1

The Rapid Prototyping Landscape

Discuss what rapid prototyping means to you.

You will:

- Define the goals that need to be achieved through a rapid prototype
- Understand the uses of the different tools and technologies used for rapid prototyping
- Compare common fabrication methods used in rapid prototyping
- Identify the benefits and fundamental issues of different fabrication processes used in rapid prototyping
- Quantify the benefits of rapid prototypes and different fabrication processes in an engineering design process

Module 2

Concept Prototype Attribute Mapping

Understand the benefits of developing a concept prototype (CP).

You will:

- Compare the characteristics of a concept prototype (CP) and the desired product (DP)
- Analyze the metrics for selecting appropriate fabrication processes
- Estimate the common costs of practical import
- Review the design attributes of user and prototype requirements
- Calculate the mass, mass moment of inertia, prototype cost, and product cost to develop a rapid prototype
- Measure the fabrication attributes and DFM consistency
- Identify the design errors in a DFF and DFM sample

Module 3

Common “Go-To” Rapid Prototyping Technologies

Identify the characteristics of manufacturing processes – serial and parallel fabrication processes.

You will compare different serial fabrication processes to determine the most cost-effective, time-saving process and list the pros and cons of:

- 3D printing fabrication
- 2D laser cutting fabrication
- 2D abrasive waterjet cutting fabrication
- 3D machining CNC milling fabrication
- 3D machining CNC turning fabrication

Module 4

Traditional Fabrication Processes

Compare the different parallel fabrication processes, such as the serial and parallel fabrication processes, to determine the most cost-effective and time-saving process in a given scenario.

You will also list the pros and cons of:

- Silicone molding fabrication
- Low-temp metal fabrication
- Thermoforming fabrication
- Sheet metal fabrication

Module 5

Smart Design, Fabrication, and Performance Assessment Decisions

Illustrate the general framework for creating a basic prototyping plan.

You will:

- Identify the possible constraints in developing a prototype
- Establish nontechnical and technical goals and requirements for a prototype
- Link the DFM and geometry of a concept prototype and the desired product
- Compare the risks and benefits of a prototype
- Devise fabrication decisions to define the strategies for creating a prototype and assessing the prototype

Module 6

Designing Concept Prototypes (CP) That Emulate Desired Products (DP)

Develop a hypothesis for a desired product.

You will:

- Analyze the attributes of a concept prototype based on a hypothesis
- Test a virtual concept prototype
- Identify the pros and cons of a virtual concept prototype
- Determine the characteristics of a virtual prototype and a physical prototype
- Assess the data gathered from testing a concept prototype based on a hypothesis

Module 7

Assessing the Cost and Value of Rapid Prototyping

Identify how rapid prototyping can be leveraged to reduce cost and increase quality.

You will:

- Calculate both the nontechnical and technical cost of a rapid prototype
- Evaluate the overall cost of developing a prototype
- Recognize how to optimize the cost value of a rapid prototype
- Discuss the cost value of using different processes for developing a prototype
- Assess how to optimize the cost value for a rapid prototype

Module 8

The Future of Manufacturing

Summarize the future of machine learning in designing and modeling rapid prototypes.

You will:

- Recognize the role of rapid prototyping in medical research and development into practice
- Discuss how rapid prototyping can be leveraged to enable advances for human health and welfare in developing countries
- Identify how 3D printing is used in the rapid prototype manufacturing process

Activities and Assignments



Capstone Project

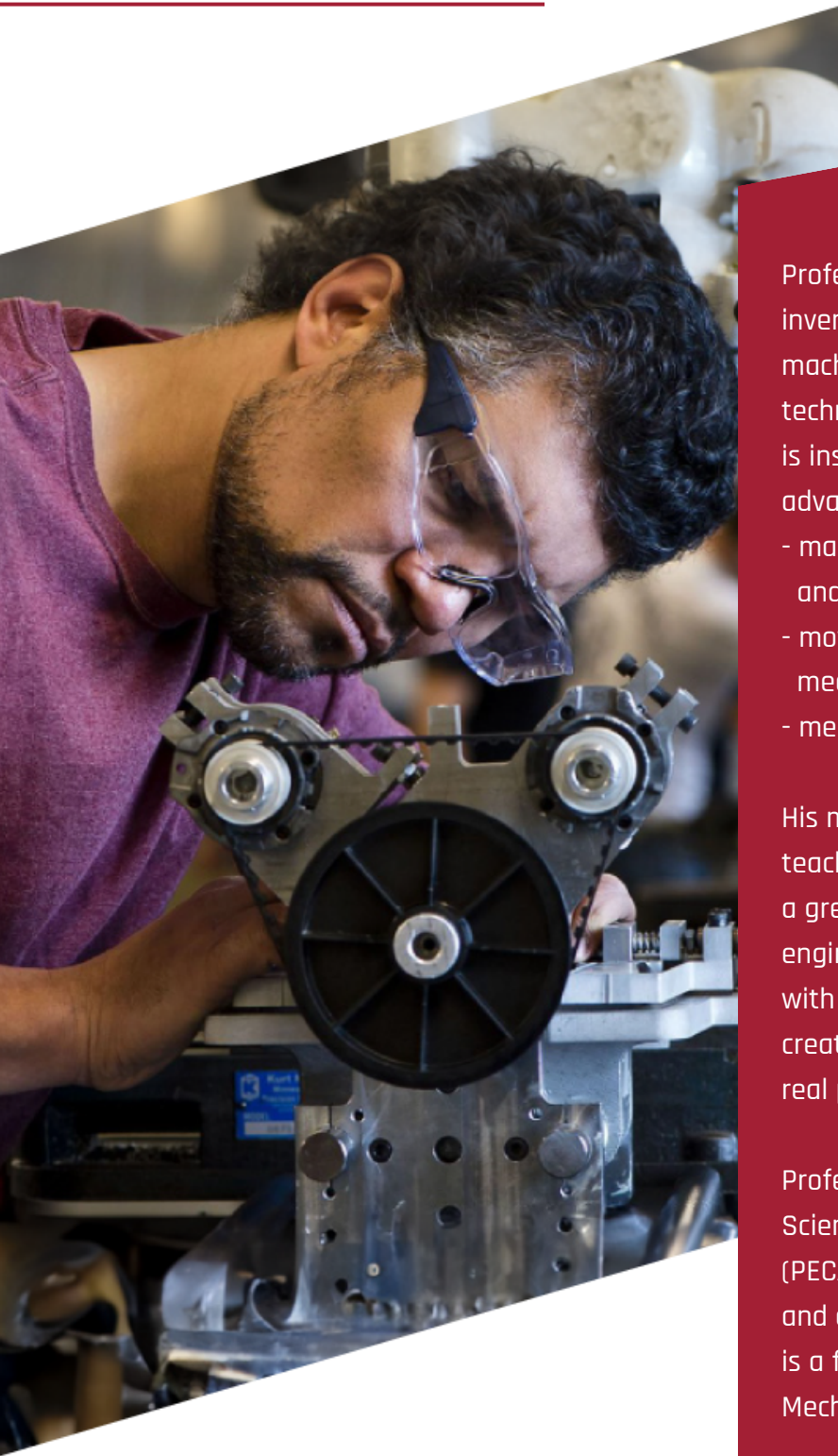
Modules 5-8 will feature a capstone assignment divided into four parts across the four modules.

Participants will receive two options of prototypes to work on: the face shield and the beam steering gyroscope for a satellite.

-  Part A: Participants will make fabrication decisions for a prototype.
-  Part B: Participants will assess the data results for the different processes used.
-  Part C: Participants will conduct a cost analysis to determine the most efficient process to use to build the prototype.
-  Part D: Participants will create a presentation/proposal based on the information gathered in Parts A-C.

Each assignment will be Level 3 graded and will have custom feedback.

Program Faculty



PROFESSOR MARTIN CULPEPPER

Professor of Mechanical Engineering at MIT

Professor Culpepper's research concentrates on the invention, design, and fabrication of high-performance machine systems. He focuses on inventing new technologies in areas where conventional technology is insufficient or impracticable, in addition to advances in machines/machine systems that:

- make (e.g., manufacturing, rapid prototyping, and fixturing);
- move (e.g., robotics, motion stages, and mechanisms); or
- measure (e.g., instrumentation and telescopes).

His mentoring (graduate students and undergraduate teaching) is based on the philosophy that becoming a great engineer means learning to meld invention, engineering common sense, and engineering science with practical fabrication/measurement skills to create parts/products/systems that solve real problems.

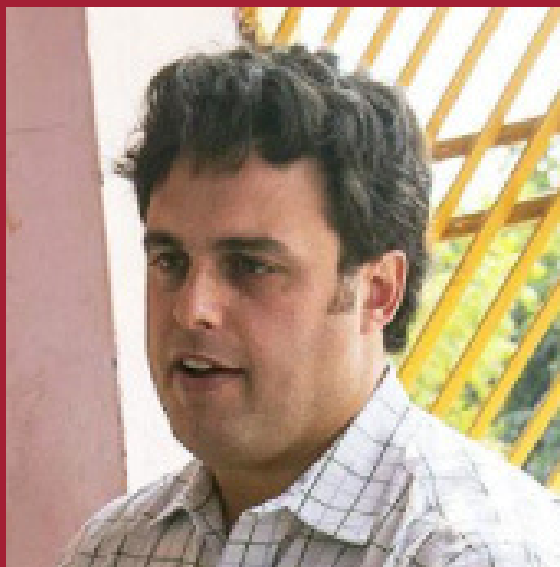
Professor Culpepper is a recipient of the National Science Foundation Presidential Early Career Award (PECASE), two R&D 100 Awards, a TR100 Award, and a Joel and Ruth Spira Teaching Award, and he is a fellow of the American Society of Mechanical Engineers.

Guest Speakers



FAEZ AHMAD

Assistant Professor, MIT



AMOS WINTER

Ratan N. Tata Career Development
Professor; Associate Professor



JOHN HART

Professor of Mechanical
Engineering; Director, Laboratory
for Manufacturing and Productivity
(LMP); Meche Maker Czar



AMOS WINTER

Director, Institute for Medical
Engineering and Science, MIT Edward J.
Poitras Professor in Medical Engineering
and Science, Massachusetts Institute of
Technology Professor of Medicine,
Harvard Medical School Senior
Attending Physician, Brigham and
Women's Hospital

Certificate

Get recognized. Upon successful completion of this program, MIT xPRO grants a certificate of completion to participants and four Continuing Education Units (CEUs). This program is graded as a pass or fail; participants must receive 75% to pass and obtain the certificate of completion.

After successful completion of the program, your verified digital certificate will be emailed to you, at no additional cost, with the name you used when registering for the program. All certificate images are for illustrative purposes only and may be subject to change at the discretion of MIT.



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